

- Description

This module manages a smart air purifier system, ensuring that:

1. The system continuously monitors the air quality index (AQI).
2. If the AQI exceeds a predefined threshold, the purifier fan speed is increased to improve air quality.
3. If the AQI is below the threshold, the purifier operates at a normal speed.
4. If the AQI reaches an extremely high level, an alert is triggered to notify users of hazardous air conditions.

This system helps maintain clean indoor air by adjusting purification levels based on real-time air quality.

- Specification

module SmartAirPurifierControl

var currentAQI, normalAQIThreshold, hazardousAQIThreshold: real

var fanSpeed: integer

process updateAQI(newAQI: real) returns (currentAQI: real, fanSpeed: integer)

functional scenario form:

currentAQI >= hazardousAQIThreshold && fanSpeed = 2, output "ALERT: Hazardous air quality (" + currentAQI + "). Fan speed set to HIGH.", Fan speed set to HIGH; hazard alert issued.

||

currentAQI < hazardousAQIThreshold && fanSpeed = 2, output "WARNING: Poor air quality (" + currentAQI + "). Fan speed increased.", Fan speed increased.

||

currentAQI <= normalAQIThreshold && fanSpeed = 1, output "Air quality is good (" + currentAQI + "). Operating at normal speed.", Fan speed remains normal.

end process

end module

- Java Code

```
package code;
public class c111_SmartAirPurifier {
    private double normalAQIThreshold;
    private double hazardousAQIThreshold;
    private double currentAQI;
    private int fanSpeed;
    public c111_SmartAirPurifier(double normalAQIThreshold, double hazardousAQIThreshold, double initialAQI) {
        this.normalAQIThreshold = normalAQIThreshold;
        this.hazardousAQIThreshold = hazardousAQIThreshold;
        this.currentAQI = initialAQI;
        this.fanSpeed = 1;
    }
}
```

```

    public void updateAQI(double newAQI) {
        currentAQI = newAQI;
        adjustFanSpeed();
        System.out.printf("Current AQI: %.2f, Fan Speed: %d\n", current
AQI, fanSpeed);
        if (currentAQI >= hazardousAQIThreshold) {
            System.out.println("ALERT: Hazardous air quality detected!
Take protective measures.");
        } else if (currentAQI > normalAQIThreshold) {
            System.out.println("WARNING: Poor air quality. Increasing f
an speed.");
        } else {
            System.out.println("Air quality is good. Operating at norma
l speed.");
        }
    }
    private void adjustFanSpeed() {
        if (currentAQI >= hazardousAQIThreshold) {
            fanSpeed = 2;
        } else if (currentAQI > normalAQIThreshold) {
            fanSpeed = 2;
        } else {
            fanSpeed = 1;
        }
    }
    public double getCurrentAQI() {
        return currentAQI;
    }
    public int getFanSpeed() {
        return fanSpeed;
    }
    public static void main(String[] args) {
        c111_SmartAirPurifier purifier = new c111_SmartAirPurifier(100.0,
200.0, 80.0);
        // testcase-VT:
        //     purifier.updateAQI(90.0);
        //     purifier.updateAQI(150.0);
        //     purifier.updateAQI(220.0);

        // testcase-FT:
        //     purifier.updateAQI(5.639821599568049E306);
        //     purifier.updateAQI(0.0);
        //     purifier.updateAQI(-1.7976931348623157E308);

```

```

        // testcase-Z3:
        purifier.updateAQI(90.0);
        purifier.updateAQI(150.0);
        purifier.updateAQI(220.0);

        // testcase-UVT:
        purifier.updateAQI(90.0);
        purifier.updateAQI(150.0);
        purifier.updateAQI(220.0);
    }
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	1.0 %	8	834	842
✓ src	1.0 %	8	834	842
✓ code	1.0 %	8	834	842
✓ c111_SmartAirPurifier.java	100.0 %	8	0	8
✓ c111_SmartAirPurifier	100.0 %	8	0	8
● updateAQI(double)	100.0 %	4	0	4
■ adjustFanSpeed()	100.0 %	4	0	4
● c111_SmartAirPurifier(double, d		0	0	0
\$ main(String[])		0	0	0

3. Vibration testing instruction coverage: 94.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.5 %	103	21,946	22,049
✓ src	0.5 %	103	21,946	22,049
✓ code	0.5 %	103	21,946	22,049
✓ c111_SmartAirPurifier.java	94.5 %	103	6	109
✓ c111_SmartAirPurifier	94.5 %	103	6	109
● updateAQI(double)	100.0 %	47	0	47
■ adjustFanSpeed()	100.0 %	24	0	24
● c111_SmartAirPurifier(double, d	100.0 %	15	0	15
\$ main(String[])	100.0 %	17	0	17

4. Fuzz testing branch coverage: 75%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.7 %	6	836	842
✓ src	0.7 %	6	836	842
✓ code	0.7 %	6	836	842
✓ c111_SmartAirPurifier.java	75.0 %	6	2	8
✓ c111_SmartAirPurifier	75.0 %	6	2	8
● updateAQI(double)	75.0 %	3	1	4
■ adjustFanSpeed()	75.0 %	3	1	4
● c111_SmartAirPurifier(double, d		0	0	0
\$ main(String[])		0	0	0

5. Fuzz testing instruction coverage: 87.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	95	21,953	22,048
▼ src	0.4 %	95	21,953	22,048
▼ code	0.4 %	95	21,953	22,048
▼ c111_SmartAirPurifier.java	87.2 %	95	14	109
▼ c111_SmartAirPurifier	87.2 %	95	14	109
● updateAQI(double)	91.5 %	43	4	47
■ adjustFanSpeed()	83.3 %	20	4	24
● c111_SmartAirPurifier(double, double)	100.0 %	15	0	15
● main(String[])	100.0 %	17	0	17

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	832	840
▼ src	1.0 %	8	832	840
▼ code	1.0 %	8	832	840
▼ c111_SmartAirPurifier.java	100.0 %	8	0	8
▼ c111_SmartAirPurifier	100.0 %	8	0	8
■ adjustFanSpeed()	100.0 %	4	0	4
● updateAQI(double)	100.0 %	4	0	4
● main(String[])		0	0	0
● c111_SmartAirPurifier(double, double)		0	0	0

7. Z3 instruction coverage: 94.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	103	22,926	23,029
▼ src	0.4 %	103	22,926	23,029
▼ code	0.4 %	103	22,926	23,029
▼ c111_SmartAirPurifier.java	94.5 %	103	6	109
▼ c111_SmartAirPurifier	94.5 %	103	6	109
● main(String[])	100.0 %	17	0	17
● c111_SmartAirPurifier(double, double)	100.0 %	15	0	15
■ adjustFanSpeed()	100.0 %	24	0	24
● updateAQI(double)	100.0 %	47	0	47

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	832	840
▼ src	1.0 %	8	832	840
▼ code	1.0 %	8	832	840
▼ c111_SmartAirPurifier.java	100.0 %	8	0	8
▼ c111_SmartAirPurifier	100.0 %	8	0	8
■ adjustFanSpeed()	100.0 %	4	0	4
● updateAQI(double)	100.0 %	4	0	4
● main(String[])		0	0	0
● c111_SmartAirPurifier(double, double)		0	0	0

9. Unimproved V-method instruction coverage: 94.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	103	22,926	23,029
▼ src	0.4 %	103	22,926	23,029
▼ code	0.4 %	103	22,926	23,029
▼ c111_SmartAirPurifier.java	94.5 %	103	6	109
▼ c111_SmartAirPurifier	94.5 %	103	6	109
● main(String[])	100.0 %	17	0	17
● c111_SmartAirPurifier(double, double)	100.0 %	15	0	15
■ adjustFanSpeed()	100.0 %	24	0	24
● updateAQI(double)	100.0 %	47	0	47